

Jayesh Bhadane

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EXPERIENCE

Skan AI – Machine Learning Engineer

Dec 2025 – Present

- Engineering an Agentic AI platform for an insurance client that automates 15+ manual workflow steps using LLM-driven agents (Claude, OpenAI, Gemini) on Google ADK, cutting per-case processing time by ~70-80%.
- Built a full-stack monitoring interface in Svelte and FastAPI with real-time workflow visualization, task-level status tracking, playbook management, and execution insights, giving operators live visibility into agent runs and faster failure resolution.
- Developed a WebSocket-based log streaming pipeline that bridged agent-emitted logs into the backend broadcast layer, powering real-time UI updates and significantly improving debuggability across multi-step agentic workflows.
- Containerized all platform components using Docker in a microservices architecture and deployed a secure authenticated reverse proxy on GCP, preventing unauthorized model access and enabling isolated agent deployments.

Aurora Engineering – Data Scientist

Aug 2024 – Dec 2025

- Built deep learning models to recover missing values in NASA's MMS 4D satellite telemetry data (~17 million data points), achieving R^2 of 0.86 on held-out data.
- Developed binary masking strategies to simulate real gap patterns in satellite telemetry, enabling robust model training on incomplete multidimensional data.
- Built recovery models for 3 target variables in the MMS dataset, each requiring independent pipelines.
- Developed algorithms to compute pitch angles across channels for interpolation, validated against reference datasets.
- Automated data acquisition and moments calculation pipelines for thousands of NASA CDF files, reducing manual processing overhead.

C5i – Sr. Data Scientist

Oct 2022 – Aug 2023

- Built behavioral and attitudinal segmentation models in Python to identify high-value user cohorts and enable personalized marketing strategies based on conversion potential.
- Validated targeted marketing strategies through A/B testing and structured experiments, achieving a 22% lift in conversion rate and measurable usage gains among priority segments.
- Applied classification, regression models and statistical modeling to perform key driver analysis, identifying high-leverage drivers or user growth that increased user engagement by 37%.
- Forecasted user volume using time series forecasting model (Prophet) informed by Granger causality, optimizing acquisition strategy and delivering \$14M in incremental revenue.
- Built a Bayesian marketing mix model using PyMC to evaluate new channel effectiveness, enabling optimized spend allocation and generating \$43.5M in incremental revenue.
- Defined and tracked key performance metrics (CPA, LTV, CAC, conversion rates) in Tableau, delivering insights that guided product adoption and engagement.

C5i – Data Scientist

Oct 2020 – Sep 2022

- Analyzed business metrics through funnel analysis in Python and SAS to identify campaign adoption drivers, executing growth strategies that generated 100k+ additional consumer interactions in 3 months.
- Leveraged factor analysis to extract latent variables and refine segmentation models, enabling more targeted marketing strategies and personalization.
- Applied CHAID modeling to clinical data to uncover patterns influencing CAR-T therapy uptake, supporting data-informed outreach efforts.
- Performed univariate and multivariate analysis to uncover trends and insights from complex datasets using statistical and data visualization techniques.
- Automated performance reporting in Python, reducing data aggregation time from hours to minutes.

EDUCATION

MS, Data Science | University of Connecticut, Storrs, USA | GPA: 3.78/4.0

Aug 2023 - Dec 2024

Bachelor of Engineering, Computer Engineering | University of Mumbai, India | GPA: 3.0/4.0

July 2016 – Oct 2020

TECHNICAL SKILLS

- **ML & Statistical Skills:** Logistic/Linear Regression, Clustering, GLMs, Decision Trees, Dimensionality Reduction, Deep Learning, Time Series Analysis, Hypothesis Testing, A/B Testing, Bayesian Modeling, Causal Inference
- **Gen AI & LLMs:** LangChain, LangGraph, LlamaIndex, LiteLLM, Google ADK, Generative AI, NLP, Agentic AI Systems, Prompt Engineering, RAG pipelines, Tool Use / Function Calling
- **Languages & Tools:** Python, R, SQL, Spark, Docker, GCP, Azure, AWS, Airflow, Databricks, Tableau, PowerBI, IBM SPSS
- **Frameworks & Libraries:** Pandas, NumPy, Scikit-Learn, XGBoost, TensorFlow, Keras, PyTorch, StatsModels, Matplotlib, Plotly, OpenCV, Rshiny, FastAPI, Svelte